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Education

Purdue University

Doctor of Philosophy in Mechanical Engineering

Thesis: *Hierarchical Radar-Camera fusion for task-sufficient 3D object detection in autonomous navigation using minimal sensor information*

Advisor: Dr. Sohel Anwar

Indianapolis, IN
December 2026 (expected)

Purdue University

Master of Science in Mechanical Engineering

GPA: 3.94/4.00

Indianapolis, IN
December 2024

Bangladesh University of Engineering and Technology

Bachelor of Science in Mechanical Engineering

GPA: 3.76/4.00

Thesis: *Transonic Aerodynamics of an Airfoil Cascade*

Advisor: Dr. A. B. M. Toufique Hasan

Dhaka, Bangladesh
August 2017

Research Interests

Control Systems, Sensor Fusion, Computer Vision, Optimization Algorithms, Reinforcement Learning, Digital Twins, Autonomous Systems

Professional Experience

Graduate Research Assistant, **MARL Lab, Purdue University**
Indianapolis, IN

August 2022 - Present

Off-Road Semantic Segmentation Dataset and Deep Learning Model

- Developed a semantic segmentation dataset of 1,414 images through label refinement of Yamaha-CMU Off-Road Dataset and class-balancing via synthetic image augmentation.
- Designed a ResNet34-U-Net fusion architecture achieving 87% accuracy at 24 Hz with state-of-the-art generalization on YCOR, Rellis-3D, and RUGD benchmarks without fine-tuning, validating robust feature representations across diverse terrain types.

PrED: Predictive Enhancement of Detection Algorithm

- Designed a novel object detection and enhancement algorithm with Kalman Filter and Predictability Index to recover low-confidence and missed detections in data-limited scenarios.
- Delivered +11% detection accuracy and +7% MOTA score in moving camera off-road tests.

Semantic-guided Hierarchical Multi-modal sensor fusion:

- Developing a three-stage radar-camera fusion with semantic-guided proposal generation to achieve task-sufficient 3D object and obstacle zone detection using only minimal sensor information.
- Modified YOLOv8 architecture with radar and semantic BEV encoders for 3D bounding box prediction.
- Incorporating PrED with the 3D object detection architecture for temporal detection improvement and obstacle avoidance zone creation.
- Lifted terrain information from Semantic Segmentation to BEV plane for traversability of off-road autonomous vehicles.
- The architecture is expected to act as a complete perception pipeline for autonomous navigation algorithms.

Physics-based Radar Sensor Model in Unreal Engine:

- Built high-fidelity automotive radar sensor model in Unreal Engine C++ integrating physics-based signal characteristics with point cloud processing and 3D grid-DBSCAN clustering.
- Achieved 89% correlation with real-world automotive radar behavior for sim-to-real control validation.

Satellite Imagery DNN for Dengue Prediction

- Built a deep neural network for spatial imagery analysis achieving 85% precision on noisy real-world data for outbreak forecasting.

DoD-Funded Autonomous Vehicle Collaboration

- Collaborated with 10+ multidisciplinary researchers to deliver ROS2 autonomy stack modules through systematic evaluation and iterative improvement.

Lab Instructor, Mechanical Engineering, Purdue University
Indianapolis, IN

January 2026 – Present

- Leading weekly hands-on experiments for 59 students across 3 sections, reinforcing core components in material testing, heat transfer analysis, and dynamic systems.
- Supervising execution of 9 laboratory experiments, guiding students through experimental design, data acquisition, and technical reporting.

Instructor, ID Tech
Seattle, WA

May 2025 – August 2025

- Delivered Python, AI and ML curriculum to 35 teens (ages 13-17) across Tech Camp sessions, covering introductory coding through basic neural networks.
- Led 2-week NVIDIA-sponsored overnight AI/ML Academy, teaching deep learning fundamentals and hands-on model deployment to pre-college participants.
- Mentored teen-led Computer Vision projects using NVIDIA Jetson Orin Nano, guiding edge deployment of object detection models from prototype to demo-ready applications.

Graduate Teaching Assistant, Mechanical Engineering, Purdue University
Indianapolis, IN

August 2024 – December 2025

- Delivered guest lectures on feedback control systems and stability analysis to 50+ students.
- Provided detailed solutions for complex control problems ranging from root locus, bode plot, state-space method and controller and compensator design.
- Facilitated office hours and problem-solving sessions and assisted students in debugging MATLAB/Simulink models.

CM Digital Transformation Lead and Assistant Quality Manager, Unilever
Dhaka, Bangladesh

April 2018 - July 2022

- Led automation initiatives across 3 plants, developing production monitoring systems with sensor integration and real-time analytics, reducing inefficiencies by 40%. Deployed predictive maintenance systems earning 2x Supply Chain Director's Awards.

Lecturer, Sonargaon University
Dhaka, Bangladesh

January 2018 - March 2018

- Designed and taught courses to undergraduate students on *Control Systems*, *Elements of Mechanical Engineering*, and *Elements of Machine Design*.

Honors and Awards

University Fellowship, **Purdue School of Engineering and Technology, IUPUI**
Technical Scholarship, **BUET**
University Merit Scholarship, **BUET**
Dean's List Award, **BUET**

2022
2013-2016
2014, 2015
2015

Publications

- **Rahi, A.**, Wasfy, H., Wasfy, T., & Anwar, S., *Improved Object Detection and Tracking with Camera in Motion Using PrED: Predictive Enhancement of Detection*, Automation, 6(4), (2025), 77.
- Khan, M. N., **Rahi, A.**, Hasan, M. A., & Anwar, S., *Decision-Level Multi-Sensor Fusion to Improve Limitations of Single-Camera-Based CNN Classification in Precision Farming: Application in Weed Detection*, Computation, 13(7), (2025), 174.
- **Rahi, A.**, Cardoza, C. O., Vemulapalli, S. S. T., El-Mounayri, H., et al., *Modeling of an Automotive Radar Utilizing Grid-DBSCAN and SNR Characteristics of Virtual Objects in Unreal Engine*, IEEE Access, 12, (2024), 127836-127845.
- **Rahi, A.**, Elgeoushy, O., Syed, S. H., El-Mounayri, H., et al., *Deep Semantic Segmentation for Identifying Traversable Terrain in Off-Road Autonomous Driving*, IEEE Access, 12, (2024), 162977-162989.
- Khan, M. N., **Rahi, A.**, Rajendran, V. P., Hasan, M. A., & Anwar, S., *Real-time Crop Row Detection Using Computer Vision—Application in Agricultural Robots*, Frontiers in Artificial Intelligence, 7, (2024), 1435686.
- Parkar, O., Snyder, B., **Rahi, A.**, & Anwar, S., *Modified Particle Swarm Optimization Based Powertrain Energy Management for Range Extended Electric Vehicle*, Energies, 16(13), (2023), 5082.

Conference Proceedings

- **Rahi, A.**, Cardoza, C. O., Vemulapalli, S. S. T., Wasfy, T., & Anwar, S., *Modeling of Automotive Radar Sensor in Unreal Engine for Autonomous Vehicle Simulation*, ASME International Mechanical Engineering Congress and Exposition, (2023), 87639.
- Vemulapalli, S. S. T., **Rahi, A.**, Mullins, S. D., Wasfy, H., et al., *IR Sensor Modeling in Unreal Engine for Autonomous Vehicle Applications*, ASME International Mechanical Engineering Congress and Exposition, (2023), 87639.
- Barua, S., **Rahi, A.**, Niloy, R. A., & Tahmid, S., *Numerical Analysis of Shock Oscillation Around a Biconvex Circular Arc Airfoil by Incorporating a Bump on It in a Channel*, International Conference on Mechanical Industrial and Materials Engineering, (2017).
- Mahmud, Z., Azad, R. I., Niloy, R. A., **Rahi, A.**, Tahmid, S., *Observation of Unsteady Shock Behavior around a Biconvex Circular Arc Airfoil with Circular Cavities on it in a channel*, International Conference on Mechanical Industrial and Materials Engineering, (2017).
- Fattah, S. A., Haider, M. Z., Chowdhury, D., Sarkar, M., Chowdhury, R. I., Islam, M. S., ... **Rahi, A.**, *An Aerial Landmine Detection System with Dynamic Path and Explosion Mode Identification Features*, 2016 IEEE Global Humanitarian Technology Conference (GHTC), (2016), 745-752.